

HEMAKUMAR P

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[Portfolio Website](#) | [LinkedIn Profile](#)

PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering student with hands-on experience in embedded systems, microcontroller programming, and hardware-software integration. Skilled in C/C++, Python, and RTL/HDL development with a strong track record of designing cost-effective smart IoT hardware solutions and AI-integrated devices.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering

Prince Shri Venkateshwara Padmavathy Engineering College, Chennai

Sep 2023 – Expected Graduation: July 2027

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java

Hardware & HDL: RTL, HDL (Verilog/VHDL), Arduino, Microcontrollers, Sensors Integration

Domains & Expertise: Embedded Systems, IoT Hardware Development, Circuit Design

Soft Skills & Languages: Technical Problem Solving, Team Leadership, English, Tamil

KEY PROJECTS

Smart Pillow (AI & Sensor-Integrated Health System)

- Engineered an intelligent pillow solution integrating temperature control, ambient sensors, and AI logic to monitor and improve sleep quality.
- Designed the system architecture to detect sleep metrics and automatically adjust parameters to mitigate sleep-related health risks.

Fuel Level Theft Detection System

- Developed an automated fuel monitoring system targeted at commercial transit and rest stops to prevent covert fuel theft.
- Integrated continuous fluid sensors with real-time alert mechanisms to preserve driver security and fleet operational efficiency.

IoT Smart Fan Retrofit

- Transformed traditional standing fans into automated IoT-enabled smart devices using Arduino microcontrollers and relay modules.
- Achieved budget-friendly environmental sensing and automated speed controls to maximize energy efficiency.

CERTIFICATIONS & AFFILIATIONS

- NPTEL Online Certification: Programming in Java
 - LEAP – Reverse Engineering & Arduino Bootcamp | 12 Sep 2023 – 15 Sep 2023
 - LPB01 – Reverse Engineering and LPB02 – Arduino Bootcamps | Prince Shri Venkateshwara Padmavathy Engineering College, Chennai
- Student Member – Indian Society for Technical Education (ISTE)
- Active Volunteer – National Service Scheme (NSS) Unit